

VU'S COVER-IDEAL LINEARITY QUESTION THROUGH TWELVE NONISOLATED VERTICES

ABSTRACT. Vu asked whether $v(J(G)^t)$ is linear in t for every bipartite graph G . We prove by exact exhaustive computation that the answer is affirmative for every bipartite graph with at least one edge and at most twelve nonisolated vertices:

$$v(J(G)^t) = v(J(G)) + (t-1)\tau(G), \quad t \geq 1.$$

Vu's local formula reduces the problem to an edge condition involving maximum independent sets. We verify the stronger statement that every edge maximizing $\alpha(G - N[e])$ has an endpoint in a maximum independent set. The computation checks 465,766,775 neighborhood-multiset encodings using exact integer arithmetic. This is a finite verification; the general question remains open.

1. RESULT

Let G be a finite simple graph, let $S = \mathbb{k}[x_v : v \in V(G)]$, and let

$$J(G) = \bigcap_{uv \in E(G)} (x_u, x_v)$$

be its cover ideal. For a nonzero homogeneous ideal $I \subsetneq S$ and $P \in \text{Ass}(S/I)$, set

$$\begin{aligned} v_P(I) &= \min\{\deg f : (I : f) = P, f \in S \text{ homogeneous}\}, \\ v(I) &= \min_{P \in \text{Ass}(S/I)} v_P(I). \end{aligned}$$

Write $\tau(G)$ and $\alpha(G)$ for the vertex-cover and independence numbers.

Theorem 1. *Let G be a finite simple bipartite graph with at least one edge and at most twelve nonisolated vertices. Then, for every integer $t \geq 1$,*

$$v(J(G)^t) = v(J(G)) + (t-1)\tau(G).$$

Vu proved local linearity [3, Theorem 1.3] and global linearity for $t \geq |V(G)| - 2$ [3, Lemma 3.1], and asked whether global linearity holds for every $t \geq 1$ [3, Question 3.2]. That question is posed immediately after the report that no bipartite graph with a nonlinear v -function of the powers of its cover ideal had been found [3, remark preceding Question 3.2], so the contribution of the present note is to make that search exhaustive through twelve nonisolated vertices rather than to introduce the first such search.

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Hang and Vu proved the tree case [1, Theorem 1.2]. Theorem 1 is a finite verification of Vu's question, not a resolution for arbitrary bipartite graphs.

2. REDUCTION TO AN EDGE CONDITION

We first dispose of isolated vertices.

Lemma 2. *Let R be a standard graded polynomial ring, let $I \subseteq R$ be homogeneous, and put $T = R[y_1, \dots, y_r]$. Then*

$$\text{Ass}_T(T/IT) = \{QT : Q \in \text{Ass}_R(R/I)\},$$

and $v_{PT}(IT) = v_P(I)$ for every $P \in \text{Ass}_R(R/I)$.

Proof. The associated-prime identity follows from flat base change. The inequality $v_{PT}(IT) \leq v_P(I)$ is immediate. Conversely, let $F = \sum_{\mathbf{a}} f_{\mathbf{a}} y^{\mathbf{a}}$ be a homogeneous witness for PT . Coefficient comparison gives

$$P = (IT : F) \cap R = \bigcap_{f_{\mathbf{a}} \neq 0} (I : f_{\mathbf{a}}), \quad P \subseteq (I : f_{\mathbf{a}}).$$

Modulo P , a finite intersection of nonzero ideals in the domain R/P is nonzero. Hence $(I : f_{\mathbf{a}}) = P$ for some coefficient, and $\deg f_{\mathbf{a}} \leq \deg F$. $\square$

We may therefore assume that G has no isolated vertices. For a vertex u , let $N[u]$ be its closed neighborhood; for an edge $e = uv$, put $N[e] = N[u] \cup N[v]$ and $P_e = (x_u, x_v)$. Since a graph is unimodular exactly when it is bipartite, standard gradedness of the vertex-cover algebra gives $J(G)^t = J(G)^{(t)}$ for every $t \geq 1$ [2, Theorem 1.1]. Thus

$$J(G)^t = \bigcap_{e \in E(G)} P_e^t.$$

The edge primes are pairwise incomparable, so each P_e is a minimal prime of $J(G)^t$ and hence lies in $\text{Ass}(S/J(G)^t)$, while the displayed intersection of P_e -primary ideals gives the reverse inclusion. Thus $\text{Ass}(S/J(G)^t) = \{P_e : e \in E(G)\}$ and

$$v(J(G)^t) = \min_{e \in E(G)} v_{P_e}(J(G)^t).$$

For $e = uv$, Vu's formula [3, Theorem 1.3 and Lemmas 2.5–2.6] becomes

- (1) $v_{P_e}(J(G)^t) = c_e + (t-1)g_e$,
- (2) $c_e = \deg(u) + \deg(v) - 2 + \tau(G - N[e])$,
- (3) $g_e = \min\{h(u), h(v)\}$, $h(u) = \deg(u) + \tau(G - N[u])$.

The common-neighbor term in Vu's general expression for c_e vanishes because G is bipartite.

Lemma 3. *Let G be bipartite without isolated vertices, let $n = |V(G)|$, and let $e = uv \in E(G)$. Then*

$$c_e = n - 2 - \alpha(G - N[e]).$$

Moreover, $h(u) \geq \tau(G)$, with equality if and only if u belongs to a maximum independent set of G , and

$$\min_{e \in E(G)} g_e = \tau(G).$$

Proof. Since $|N[e]| = \deg(u) + \deg(v)$ in a bipartite graph, equation (2) and $\tau(H) = |V(H)| - \alpha(H)$ give the first identity. The quantity $h(u)$ is the minimum size of a vertex cover avoiding u : such a cover must contain $N(u)$ and then cover $G - N[u]$. Hence $h(u) = \tau(G)$ exactly when a minimum vertex cover avoids u , equivalently when a maximum independent set contains u . Finally, choose a vertex u in a maximum independent set and an incident edge uv . Then $g_{uv} \leq h(u) = \tau(G)$, while every $g_e \geq \tau(G)$. $\square$

Define

$$\mathcal{M}(G) = \arg \max_{e \in E(G)} \alpha(G - N[e]).$$

Proposition 4 (Edge criterion). *For a bipartite graph G without isolated vertices, the following are equivalent:*

- (1) $t \mapsto v(J(G)^t)$ is affine for every integer $t \geq 1$;
- (2) $v(J(G)^t) = v(J(G)) + (t - 1)\tau(G)$ for every integer $t \geq 1$;
- (3) some edge in $\mathcal{M}(G)$ has an endpoint in a maximum independent set of G .

Proof. By (1), the global v -number is the lower envelope of finitely many lines $c_e + (t - 1)g_e$. Its value at $t = 1$ is $\min_e c_e = v(J(G))$, and its eventual slope is $\min_e g_e = \tau(G)$ by Lemma 3. A finite lower envelope is affine on all positive integers if and only if one line minimizes both its intercept and its slope: in the nontrivial direction, a line attaining the envelope for infinitely many integers agrees with the affine envelope at two points and hence identically. Lemma 3 identifies the intercept minimizers with $\mathcal{M}(G)$ and the slope minimizers with the edges having an endpoint in a maximum independent set. $\square$

3. EXACT EXHAUSTIVE VERIFICATION

Fix $n \leq 12$, choose $1 \leq q \leq \lfloor n/2 \rfloor$, and put $p = n - q$. Every bipartite graph without isolated vertices is represented, possibly more than once, by a tuple

$$(A_1, \dots, A_q), \quad \emptyset \neq A_i \subseteq [p], \quad A_1 \cup \dots \cup A_q = [p],$$

where A_i is the neighborhood of the i th vertex on the q -side, and the tuple is nondecreasing with respect to the integer value of the characteristic bitmask of A_i in $[p]$.

Let $L = [p]$, $R = [q]$, and $U(B) = \bigcup_{i \in B} A_i$ for $B \subseteq R$. For every encoded graph,

$$(4) \quad \alpha(G) = \max_{B \subseteq R} (|B| + p - |U(B)|).$$

Indeed, once the right part B of an independent set is fixed, every vertex of $L \setminus U(B)$ may be added. If $e = xi$ with $x \in L$ and $i \in R$, then

$$(5) \quad \alpha(G - N[e]) = \max_{B \subseteq R \setminus N(x)} (|B| + p - |A_i \cup U(B)|).$$

The maximizers in (4) also determine membership in maximum independent sets: $i \in R$ occurs in one if and only if some maximizing B contains i , while $x \in L$ occurs in one if and only if some maximizing B satisfies $x \notin U(B)$. Thus (4)–(5) give an exact bitset test of the condition in Proposition 4.

For fixed (p, q) , the number of tuples before imposing full union is

$$E(p, q) = \binom{2^p + q - 2}{q},$$

and inclusion–exclusion gives the retained count

$$K(p, q) = \sum_{r=0}^p (-1)^r \binom{p}{r} \binom{2^{p-r} + q - 2}{q},$$

with $\binom{a}{b} = 0$ for $a < b$. The generated counts agree with these formulas for every (p, q) .

TABLE 1. Complete neighborhood-multiset census. In no retained encoding does an edge maximizing $\alpha(G - N[e])$ fail to have an endpoint in a maximum independent set; see Proposition 5 below.

n	generated	retained
2	1	1
3	3	1
4	13	5
5	43	14
6	235	98
7	1,239	522
8	10,659	5,472
9	98,439	53,733
10	1,428,007	895,828
11	23,944,527	16,033,665
12	610,303,679	448,777,436
Total	635,786,845	465,766,775

Proposition 5. *Let G be a bipartite graph without isolated vertices and with $|V(G)| \leq 12$. Every edge in $\mathcal{M}(G)$ has an endpoint belonging to a maximum independent set of G .*

Proof. The enumeration above covers every such graph. The computation evaluates (4) and (5) using integer bit operations and checks every edge

in $\mathcal{M}(G)$. None of the 465,766,775 retained encodings violates the stated condition. $\square$

Proof of Theorem 1. Delete isolated vertices using Lemma 2. Proposition 5 supplies an edge satisfying Proposition 4(3), which yields the asserted formula. $\square$

Exact verification. Everything the computation does is fixed by the data printed above: the encoding of Section 3, the closed forms $E(p, q)$ and $K(p, q)$, the two bitset formulas (4)–(5), and the rule that reads membership in a maximum independent set off the maximizers of (4). All arithmetic is exact integer and bit-mask arithmetic, with no floating point and no randomness. The computation was re-run once independently, on a separate machine, by a reimplementaion written from that data alone, and all 53 of its checks agree with what is reported here: both columns of Table 1 and both totals come out three ways, from the closed forms, from an unbounded-knapsack dynamic program and from prefix enumeration; equations (4)–(5), the membership rule, Lemma 3 and Propositions 4 and 5 agree with brute force on all 59,846 retained encodings with $n \leq 9$; and the condition of Proposition 5 fails on 4,381 of the 28,262 labelled graphs without isolated vertices on three to six vertices once bipartiteness is dropped, while no bipartite graph among those 28,262 fails it. The 955,674 retained encodings with $n \leq 10$ were rescanned one at a time, whereas the 464,811,101 with $n \in \{11, 12\}$ were covered through the 110,249,245 orbit representatives of the transposed scan rather than individually, which leaves no graph untested, since every retained encoding of a cell (p, q) is a relabelling of the p -side of exactly one representative and $\alpha(G)$, the union of the maximum independent sets and $\mathcal{M}(G)$ are isomorphism invariants, and the two routes were compared cell by cell wherever both are affordable, including two cells with $n = 12$. Lemma 2, the gradedness $J(G)^t = J(G)^{(t)}$ of [2] and Vu's local formula (1)–(3) were taken as cited and not re-derived. The independent reimplementaion and its transcript accompany this note; the program that produced Table 1 is retained in an auxiliary archive available on request. The re-run is a second computation, not a proof.

Scope. The result above is confined to bipartite graphs with at least one edge and at most twelve nonisolated vertices. Vu's question for arbitrary bipartite graphs remains open.

REFERENCES

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